

Preliminary Investigation on Wrist Temperature Control by Wearable Peltier Armband

Koki Akiyama
koki.akiyama@iis.ise.ritsumei.ac.jp
Ritsumeikan University
Shiga, Japan

Kazuya Murao
murao@cs.ritsumei.ac.jp
Ritsumeikan University
Shiga, Japan

ABSTRACT

Due to COVID-19 pandemic, body temperature measurement in commercial facilities is performed using a non-contact method. However, if the human body can be controlled in some way to disguise body temperature, a thermometer would have difficulty detecting an entrant with a fever. In this study, we propose a method to control body temperature measured at the wrist in order to demonstrate the vulnerability of temperature measurement at the wrist. Our device lowers body temperature by cooling the upper arm, thereby cooling blood flow and reducing the intensity of infrared radiation detected by a thermometer. The implemented device was used to cool the upper arm under three different conditions. The results showed that cooling the blood flow in the upper arm can lower the body temperature at the wrist. The cooled body temperature was difficult to maintain after the end of cooling, irrespective of the cooling intensity and cooling time.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Body temperature control, Peltier element, Wearable, COVID-19

ACM Reference Format:

Koki Akiyama and Kazuya Murao. 2022. Preliminary Investigation on Wrist Temperature Control by Wearable Peltier Armband. In *Proceedings of the 2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC '22 Adjunct)*, September 11–15, 2022, Cambridge, United Kingdom. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544793.3560394>

1 INTRODUCTION

As a measure to prevent COVID-19 infection, commercial facilities such as restaurants, department stores, and theme parks measure body temperatures for visitors. The axilla, oral cavity, forehead, ear, and wrist are used as measurement sites. In particular, axillary thermometers are less susceptible to outside air and can make accurate measurements, but many thermometers are contact thermometers,

which require the measurement part to be in direct contact with the skin. Therefore, in places with many people coming and going, such as large commercial facilities, axillary temperature measurement using contact thermometers for all facility users is time-consuming. Furthermore, the cost of disinfectant solution and personnel to perform the disinfection is high, so it has not been adopted as a temperature measurement method.

In general, non-contact body temperature measurement is used in commercial facilities. Non-contact temperature measurements are made at the forehead and wrist. The forehead temperature measurement requires the user to raise the forehead hair or to take off the hat. Therefore, wrist temperature measurement, which is less burdensome on the visitors, is becoming more popular. However, non-contact thermometers have limitations and research is being conducted to confirm their accuracy [1–3, 8]. Temperature sensing at the wrist using a non-contact thermometer makes it difficult to detect an admission with a fever if the temperature at the wrist can be temporarily lowered by some form of control of the human body. Therefore, if the temporary lowering of wrist temperature becomes widespread, there is a danger that febrile people who wish to use commercial facilities will be able to use them by falsifying their own body temperatures.

This study proposes a device worn on the upper arm to cool blood flow and control body temperature measured at the wrist. The proposed method uses Peltier elements to cool the upper arm, which lowers the body temperature measured at the wrist by cooling the blood flow and weakening the infrared radiation emitted from the wrist area. Through the proposed method, the vulnerability of wrist thermometry can be presented, and it is believed that the use of appropriate body temperature measurement and the use of advanced thermometry methods can be recommended, thereby realizing a safe and secure society.

2 RELATED WORK

Peltier devices are thin plate thermoelectric conversion devices in which the application of a voltage causes a heat transfer from one side to the other side. Uehara et al. [9] reported that the small size, light weight, lack of noise and vibration, and the ability to control temperature by switching the applied voltage make it a promising application as a biological cooling device. Other studies have been carried out to improve the performance of Peltier elements [10] and to control temperature using Peltier elements [6].

Kume et al. [5] have shown that a water circulation suit is effective in suppressing the rise in body temperature against heat stroke occurring in various locations, such as during exercise and at work sites. Furthermore, the relationship between cooling capacity, cooling area and thermoregulatory response was clarified, suggesting

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp/ISWC '22 Adjunct, September 11–15, 2022, Cambridge, United Kingdom

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9423-9/22/09...\$15.00

<https://doi.org/10.1145/3544793.3560394>

Table 1: Temperature [°C] at the both wrists when cooling the upper right arm for five minutes.

	Start	1 min	2 min	3 min	4 min	5 min
Right wrist	36.4	36.2	36.1	36.1	36.1	36.1
Left wrist	36.4	36.4	36.3	36.3	36.2	36.3

that changes in blood circulation due to body cooling may influence thermoregulatory response. Luke et al. [7] reported that measuring a person's deep body temperature is important in assessing their health status. Habeeb et al. [4] validated four commercially available ambient temperature sensors and demonstrated the impact of sun exposure, urban form, physical movement, and on-body placement on the reliability of those sensors. Their investigation quantified the degree to which wearable temperature sensors are less reliable in highly urban areas and when worn in direct sunlight regardless of activity or on-body placement.

3 PROPOSED METHOD

This section describes a method to control the body temperature measured at the wrist by cooling the upper arm. The device worn by the person to be tested has Peltier elements as a cooling device in contact with the skin. When the device is turned on, the blood flow in the upper arm is cooled to control the temperature measured at the wrist.

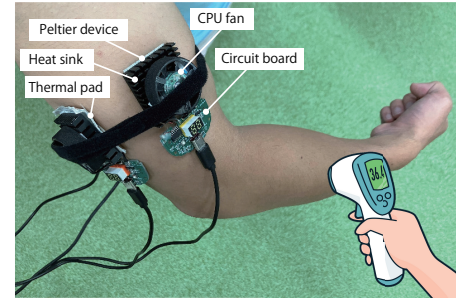
3.1 Preliminary Experiment

A preliminary experiment was conducted using an instant cold pack to verify whether cooling the blood flow in the upper arm can lower the measured body temperature at the wrist by decreasing the intensity of infrared radiation emitted from the wrist area.

3.1.1 Setup. One subject (male, right-handed, in his 20s, no fever) was asked to wear a commercially available instant cold pack (LM-802PM) on his right upper arm at the position where the proposed device was supposed to be worn. The cold pack was applied to the right upper arm for 5 minutes, and body temperature was measured every minute on both the cooled (right) and uncooled (left) wrists. Commercially available non-contact thermometers (DT-103 by ICST Corporation)¹ were used to measure body temperature. The body temperature at the uncooled wrist was compared with the body temperature measured at the cooled wrist to evaluate the decrease in body temperature due to blood flow cooling at the upper arm.

3.1.2 Results and Discussion. Table 1 shows the body temperatures measured on the left and right wrists for five minutes from the start of cooling. The temperature at the right wrist decreased from 36.4°C to 36.1°C. On the other hand, the temperature did not fall at the left wrist. Furthermore, it was found that it takes time for the body temperature to begin to drop. It was confirmed that the body temperature was lowered, suggesting that cooling of the upper arm can affect the temperature measured at the wrist.

¹https://icst.jp/whatsnew/DT-103_20201021.pdf

**Figure 1: Proposed device.**

3.2 Assumed Environment

Although refrigerants and ice, as used in the preliminary experiments, are possible methods for cooling the upper arm, it is difficult to use ice or refrigerants all the time. Therefore, the device proposed in this study is designed to cool the upper arm at all times and to control the cooling temperature. The proposed device also takes into consideration the ease with which the device user perceives the temperature and the length of the circumference of the dominant upper arm, making it possible to change the cooling temperature and accommodate a wide or narrow arm by attaching the device using Velcro tape.

3.3 Hardware

For the mounting device, a commercially available mobile Peltier cooler with temperature control function, DN-916044, was disassembled and its exterior parts were removed to reveal the Peltier device, heat sink, CPU fan, circuit board, and temperature sensor. The thermal pad was mounted by fixing it to the surface of the heat sink using heat-resistant tape. A power switch is attached to the board of the mounting device, and blood flow cooling can be started and stopped at arbitrary timing. The subject wearing the device is shown in Figure 1. The power switch is visible to the wearer of the mounting device when the device is worn, so that the blood flow cooling can be stopped immediately if the wearer feels something wrong with his/her body when using the mounting device. The cooling temperature can be adjusted from 5°C to 40°C in 1°C increments by manipulating the board. The temperature setting of the mounting device at the start of cooling inherits the last temperature setting of the previous use. The proposed device uses four mounting devices for efficient upper arm cooling, and these devices are fixed to Velcro with a glue gun.

3.4 How to Use the Device

The user attaches the proposed device so that the Peltier devices cover the blood vessels on the elbow side of the upper arm where the skin is thinner and visible. When cooling is initiated after the device is attached, blood flow is cooled and the cooled blood flows to the forearm. Therefore, in the measurement of body temperature at the wrist, temperature is measured by detecting infrared radiation emitted from the skin surface. If the temperature is low, the intensity of the infrared radiation is considered weak. In other words, the cooling of the blood flow lowers the measured body temperature

by weakening the intensity of the infrared radiation around the wrist. In this paper, we assume that the user of the proposed device hides the use of the proposed device from the temperature examiner when the user enters a commercial facility. If blood flow cooling is used to reduce body temperature, it is considered more efficient to cool the blood vessels around the wrist, i.e., the veins in the palm and back of the hand. However, there is a possibility that the temperature examiner may discover that the device is worn on the palm or back of the hand. Therefore, regardless of the four seasons, the device is attached in areas where it can be concealed by clothing, i.e. the upper arm was used because it is close to the area where body temperature is measured and because the skin is thin and it is considered easy to cool the blood flow.

4 EVALUATION

Two evaluation experiments 1 and 2 are conducted. In the experiment 1, subjects wore the proposed device to cool blood flow in their upper arms and measured the temperature at the wrists. The experiment 2 was conducted to evaluate the effects of varying the cooling intensity and cooling time using the proposed device on the body temperature measurement at the wrist, respectively.

4.1 Experiment 1

Experiment 1 was conducted to investigate the effect of blood flow cooling at the upper arm by the proposed device on body temperature measurement at the wrist.

4.1.1 Setup. We evaluated a method of controlling body temperature measurement at the wrist by cooling the upper arm of the dominant arm (right arm) using the proposed device on five subjects. The same procedure as in the preliminary experiment was used to measure body temperature. The cooling time was five minutes. In order to prevent the heat from the heating surface of the Peltier element from being transferred to the cooling surface, a heat sink and CPU fan were used to dissipate the heat, and a heat shield was attached to the mounting surface of the proposed device to prevent the subject from being burned. If the subject feels heat or pain around the cooling area, the experiment is terminated immediately. As in the preliminary experiment, subjects had no fever. The thermometer used was the same as in the preliminary experiment. The temperature setting of the proposed device was set to 5°C, which is the maximum cooling.

4.1.2 Results and discussion. The results of the experiment are shown in Table 2. The results show that the body temperature measured at the wrist of the right arm where upper arm cooling was performed varied from person to person, and for the four subjects other than subject A, the body temperature gradually decreased after the start of the measurement. From the results of Subject A, Subject A failed to control body temperature when accurate cooling was not achieved due to differences in arm thickness and misalignment of the cooling site. In subject B~E, the ability to cool the upper arm in the correct position resulted in a lower blood temperature, which may have lowered the body temperature measured at the wrist. The results show that the proposed device can produce the same effect as the medical cooling packs used in the preliminary experiments. As in the preliminary experiment, we also confirmed

Table 2: Wrist temperature [°C] when the proposed device is worn on the right arm to cool blood flow.

Subject	Wrist	Start	1 min	2 min	3 min	4 min	5 min
A	R	36.1	36.2	36.1	36.1	36.1	36.1
	L	36.4	36.4	36.3	36.3	36.2	36.3
B	R	36.4	36.3	36.1	36.0	36.0	36.2
	L	36.7	36.5	36.5	36.4	36.4	36.4
C	R	36.5	36.5	36.4	36.3	36.2	36.2
	L	36.6	36.5	36.5	36.5	36.5	36.5
D	R	36.3	36.0	36.0	35.9	36.0	35.7
	L	36.3	36.3	36.0	36.2	36.2	36.2
E	R	36.5	36.4	36.2	36.1	36.1	36.0
	L	36.4	36.4	36.3	36.4	36.4	36.5

that the decrease in body temperature was not constant, that the range of the decrease varied, and that there was a time lag before the onset of the decrease in body temperature.

4.2 Experiment 2

Experiment 2 was conducted to investigate the effect of varying the cooling intensity and cooling time on the body temperature measurement at the wrist, respectively.

4.2.1 Setup. Five subjects (F–J, only J is female, 20–23 years old, all right-handed), different from the subjects in the experiment 1, were asked to fix four of the proposed devices to Velcro and wear them closely on the right upper arm as in the experiment 1, and the difference in the range of body temperature decrease according to cooling intensity and cooling time was measured. Experiments were conducted in the following three environments

- The right arm is cooled with the proposed device for five minutes, then the device is turned off for one minute. The device is set to 5°C. This setting is to see how the cooled temperature is maintained after using the proposed device.
- The device is set to 15°C and the arm is cooled for five minutes, then the device is turned off for one minute. This setting is to see the effect of cooling the body with a lower cooling intensity
- The device is set at 5°C, and the arm is cooled for ten minutes, then the device is turned off for one minute. This setting is to see the effect of cooling the body for longer time.

In experiment 2, five subjects were tested once each under three different conditions. The same experiments as in Evaluation Experiment 1 were conducted, followed by experiments in which the cooling intensity was reduced, and finally, experiments in which the cooling time was increased. To investigate the maintenance of the cooling body temperature at the end of cooling, the body temperature one minute after the end of cooling was also measured. In actual use, it is unlikely that the body temperature would be cooled continuously. Therefore, an interval of 10 to 30 minutes was provided between experiments for one subject, and the experiment was resumed when the skin temperature reached the same level as that of the upper arm on the side that was not being cooled. The temperature setting in Evaluation Experiment 1 was set to a high setting. The low set temperature is set at 15°C.

Table 3: Wrist temperature [$^{\circ}\text{C}$] when the device is set to 5°C and cooled the arm for five minutes, then stopped cooling for one minute.

Subject	Wrist	Start	1 min	2 min	3 min	4 min	5 min	6 min
F	R	36.8	36.5	36.5	36.5	36.6	36.3	36.7
	L	36.7	36.7	36.9	36.9	36.5	36.6	36.6
G	R	36.7	36.6	36.6	36.5	36.4	36.3	36.8
	L	36.8	36.8	37.0	36.9	36.9	37.0	36.9
H	R	35.8	35.6	35.6	35.6	35.5	35.5	35.7
	L	35.9	35.8	35.7	35.7	35.7	35.7	35.7
I	R	36.4	36.1	36.1	35.9	35.9	35.7	36.2
	L	36.3	36.2	36.0	36.2	36.2	36.2	36.3
J	R	35.7	34.6	34.6	34.3	34.1	34.3	35.1
	L	35.6	35.7	35.6	35.3	35.3	35.3	35.6

Table 4: Wrist temperature [$^{\circ}\text{C}$] when the device is set to 15°C and cooled the arm for five minutes, then stopped cooling for one minute.

Subject	Wrist	start	1 min	2 min	3 min	4 min	5 min	6 min
F	R	36.7	36.8	36.5	36.5	36.4	36.5	36.7
	L	36.7	36.6	36.9	36.7	36.8	36.7	36.7
G	R	36.7	36.5	36.5	36.3	36.4	36.3	36.6
	L	36.8	36.8	36.8	36.8	36.8	36.8	36.9
H	R	36.0	36.0	35.9	36.0	36.0	35.9	36.2
	L	36.0	36.1	36.1	36.1	36.1	36.0	36.0
I	R	36.4	36.3	36.2	36.2	36.1	36.0	36.3
	L	36.3	36.4	36.3	36.3	36.4	36.4	36.3
J	R	36.3	36.0	35.9	35.7	35.8	35.6	36.0
	L	36.3	36.2	36.1	36.0	36.1	36.2	36.1

Table 5: Wrist temperature [$^{\circ}\text{C}$] when the device is set to 5°C and cooled the arm for ten minutes, then stopped cooling for one minute.

Subject	Wrist	Start	1 min	2 min	3 min	4 min	5 min	6 min	7 min	8 min	9 min	10 min	11 min
F	R	36.1	36.1	36.3	36.5	36.5	36.5	36.5	36.4	36.5	36.2	36.3	36.6
	L	36.5	36.6	36.6	36.7	36.6	36.7	36.6	36.7	36.7	36.7	36.6	36.6
G	R	36.7	36.6	36.6	36.5	36.4	36.3	36.3	36.3	36.4	36.2	36.3	36.6
	L	36.9	37.0	37.0	36.9	36.9	36.8	36.8	36.9	36.8	36.8	36.9	36.8
H	R	35.7	35.5	35.5	35.6	35.7	35.6	35.5	35.3	35.2	35.3	35.3	35.6
	L	35.7	35.7	35.7	35.7	35.7	35.7	35.7	35.7	35.6	35.7	35.6	35.8
I	R	36.4	36.3	36.2	36.2	36.1	36.0	36.1	36.0	36.1	36.1	36.0	36.2
	L	36.3	36.4	36.3	36.3	36.4	36.4	36.3	36.3	36.4	36.3	36.3	36.4
J	R	36.0	35.4	35.4	35.5	35.4	35.3	35.3	35.4	35.4	35.4	35.4	35.8
	L	36.0	36.2	36.1	36.0	36.1	36.2	36.3	36.3	36.2	36.3	36.3	36.2

4.2.2 Results and discussion. The results of an experiment in which the upper arm was cooled for five minutes at a set temperature of 5°C are shown in Table 3. The times in the table are the elapsed time from the start of cooling. The results of the experiments conducted with reduced upper arm cooling intensity (15°C) and longer upper arm cooling time (ten minutes) are shown in Table 4 and Table 5, respectively. The results of the experiment showed that when the upper arm cooling was carried out at 15°C , the body temperatures of all four subjects except subject H fell, but in many cases the fall

was less than that at 5°C , suggesting that the body temperature measured at the wrist can be reduced more by increasing the cooling intensity. The body temperature of the female subject J was lower than that of the other male subjects. In the case of Subject J, the lower muscle mass and thinner skin of her body caused a greater drop in body temperature and maintenance of a cooled body temperature than in the other subjects. From this result, muscle mass influences the range of temperature decrease.

When upper arm cooling was performed with a longer cooling time, it was found that the body temperature did not constantly decrease during cooling, but stopped and stagnated around the body temperature which is considered to be the limit of body temperature decrease, although there were individual differences between subjects when compared with the experimental results with a shorter cooling time. The cooled body temperature was not maintained when the device was set to 15°C and returned to the body temperature one min after the end of cooling. However, in the experiment with 10-min cooling time, the body temperature measured one minute after the end of cooling was closer to the body temperature measured before the start of cooling. The cooling body temperature was considered to be maintained compared to the experiment with a shorter cooling time.

In this paper, four mounting devices were used to cool the upper arm of the dominant arm. However, as a result of not fully taking into account the differences in the thickness of the upper arm of each individual, it is thought that the devices failed to lower the body temperature because they did not provide adequate upper arm cooling. For more accurate upper arm cooling, the device used must be adapted to the thickness of the upper arm. The misalignment of the cooling sites may have resulted in a failure of the upper arm cooling to reduce the measured body temperature at the wrist in each experiment. Based on these results, it is necessary to conduct more detailed experiments by increasing or decreasing the number of devices used and by employing subjects with different muscle mass.

5 CONCLUSION

In this paper, in order to demonstrate the vulnerability of using the wrist for temperature measurement at the time of admission in commercial facilities, and to recommend the use of temperature measurement in appropriate areas and advanced temperature measurement methods, we proposed a method to lower the body temperature measured by a non-contact thermometer by cooling the upper arm using a Peltier element. Evaluation experiments confirmed that body temperature was successfully lowered for nine of ten subjects. We will downsize the hardware and improve it to get arbitrary body temperatures.

ACKNOWLEDGMENTS

This research was supported in part by the Japan Science and Technology Agency, PRESTO grant number JPMJPR1937, Japan.

REFERENCES

- [1] Scott D. Adams, Tracey K. Bucknall, and Abbas Z. Kouzani. 2020. A Study on the Agreement of Body Temperatures Measured by Infrared Cameras and Oral Thermometry. <https://doi.org/10.21203/rs.3.rs-123101/v1>
- [2] Nishant Aggarwal, Mohil Garg, Vignesh Dwarakanathan, Nitesh Gautam, Swasthi S Kumar, Ranveer Singh Jadon, Mohak Gupta, and Animesh Ray. 2020. Diagnostic accuracy of non-contact infrared thermometers and thermal scanners: A systematic review and meta-analysis. *Journal of Travel Medicine* 27, 8 (2020), taaa193. <https://doi.org/10.1093/jtm/taaa193>
- [3] Giovanni Battista Dell'Isola, Elena Cosentini, Laura Canale, Giorgio Ficco, and Marco Dell'Isola. 2021. Noncontact body temperature measurement: uncertainty evaluation and screening decision rule to prevent the spread of COVID-19. *Sensors* 21, 2 (2021), 346. <https://doi.org/10.3390/s21020346>
- [4] Dana Habeeb, James Clawson, Arash Zakeresfahani, and Zebulon Holtz. 2022. Investigating and Validating On-body Temperature Sensors for Personal Heat Exposure Tracking. (2022), 1–14. <https://doi.org/10.1145/3491102.3517631>
- [5] MASASHI KUME, TESTUYA YOSHIDA, HIDEYUKI TSUNEOKA, NAOTO KIMURA, and TAKASHI ITO. 2009. RELATIONSHIP BETWEEN BODY SURFACE COOLING AREA, COOLING CAPACITY, AND THERMOREGULATORY RESPONSES WEARING WATER PURFUSED SUITS DURING EXERCISE IN HUMANS. *Japanese Journal of Physical Fitness and Sports Medicine* 58, 1 (2009), 109–122. <https://doi.org/10.7600/jspfsm.58.109>
- [6] Gianluca A. Mannella, Vincenzo La Carrubba, and Valerio Brucato. 2014. Peltier cells as temperature control elements: Experimental characterization and modeling. *Applied Thermal Engineering* 63, 1 (2014), 234–245. <https://doi.org/10.1016/j.applthermaleng.2013.10.069>
- [7] Luke N. Steck, Ephraim M. Sparrow, and John P. Abraham. 2011. Non-invasive measurement of the human core temperature. *International Journal of Heat and Mass Transfer* 54, 4 (2011), 975–982. <https://doi.org/10.1016/j.ijheatmasstransfer.2010.09.042>
- [8] Alex Buoite Stella, Paolo Manganotti, Giovanni Furlanis, Agostino Accardo, and Milo Ajevi. 2020. Return to school in the COVID-19 era: considerations for temperature measurement. *Journal of Medical Engineering & Technology* 44 (2020), 468 – 471. <https://doi.org/10.1080/03091902.2020.1822941>
- [9] Kenyu Uehara, Kentaro Miyago, Koji Mori, and Takashi Saito. 2016. Study on Model Parameters of Focal Cooling Device Using a Peltier Element for a Living Body (ASME International Mechanical Engineering Congress and Exposition, Vol. Volume 3: Biomedical and Biotechnology Engineering). https://doi.org/10.1115/IMECE2016-67387_V003T04A087
- [10] DDL Wijngaards, E Cretu, SH Kong, and RF Wolffenbuttel. 2000. Modelling of integrated Peltier elements. (2000), 652–655. null ; Conference date: 27-03-2000 Through 29-03-2000.